Zainab Iqbal

Senior AI Engineer | Multi-Agent Systems & LLM Optimization

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PROFESSIONAL SUMMARY

Senior AI Engineer with 4+ years building sophisticated language model systems and multi-agent frameworks in government technology. Specialized in retrieval-augmented generation (RAG) pipelines, fine-tuning optimization, and evaluation methodologies for complex NLP tasks. Proven ability to architect scalable AI solutions that integrate with legacy systems and modernize policy research workflows. Strong track record translating research-grade techniques into production-grade implementations.

EXPERIENCE

Senior AI Engineer | Department of Legislative Affairs, Commonwealth Technology Division **|** Jan 2024 – Present

- Architected multi-agent framework for automating bill summary generation and legislative impact analysis using GPT-4 and Claude with dynamic routing logic
- Implemented RAG pipeline integrating 50,000+ archived legislative documents and regulatory texts; achieved 89% retrieval precision on held-out evaluation set
- Fine-tuned BERT-based classifier for constituent petition categorization, improving F1 score from 0.71 baseline to 0.94 on domain-specific benchmark
- Designed comprehensive evaluation suite measuring latency (<200ms p95), factual consistency, and legal terminology accuracy across all model outputs
- Led technical strategy for integrating Anthropic API and open-source embeddings with existing government data infrastructure; published internal guidelines adopted across 3 departments
- Contributed research paper on efficient prompt caching for high-volume document processing; presented findings at ACL 2025 workshop

AI/ML Engineer | Public Sector Modernization Labs **|** Mar 2023 – Dec 2023

- Developed agentic workflow system for automated compliance verification using function-calling and structured outputs; evaluated on 2,000 synthetic regulatory scenarios with 92% accuracy
- Built end-to-end fine-tuning pipeline for adapting T5 models to government administrative language; optimized inference latency by 35% through quantization and batching strategies
- Implemented advanced RAG system combining dense and sparse retrieval; benchmarked against keyword search baseline, achieving 23% improvement in MRR@10 on internal document corpus
- Created evaluation framework measuring hallucination rates, semantic similarity, and task-specific metrics across 15 different government use cases
- Open-sourced custom evaluation utilities; repository reached 340 GitHub stars and featured in 2 community blogs

Machine Learning Engineer | Civic Data Systems Inc. **|** Jun 2022 – Feb 2023

- Engineered prompt optimization framework and chain-of-thought templates for government benefits eligibility determination; reduced token usage by 28% while maintaining 96% answer accuracy
- Built distributed fine-tuning pipeline for adapting open-source LLMs to government domain; trained on 100K+ labeled examples using LoRA and QLoRA techniques
- Developed comprehensive evaluation harness benchmarking model outputs against established policy guidelines; tracked performance across 12 quantitative metrics
- Collaborated with policy analysts on specification of reasoning requirements; translated domain requirements into structured prompts and evaluation criteria

EDUCATION

Master of Science, Computer Science | University of Pennsylvania **|** 2022

Thesis: Advanced Retrieval Techniques for Multi-Document Question Answering

Bachelor of Science, Computer Science with Mathematics Minor | Temple University **|** 2020

SKILLS & CERTIFICATIONS

Technical: Python, PyTorch, LangChain, LlamaIndex, Hugging Face Transformers, Claude/GPT-4 APIs, Fine-tuning (LoRA/QLoRA), RAG systems, Multi-agent frameworks, Prompt engineering, Evaluation methodologies, Vector databases (Pinecone, Weaviate), Cloud infrastructure (AWS, GCP)

Domain: Government technology, regulatory compliance, NLP, information retrieval, model optimization